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Data Mining

Self-Organizing Maps

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NOVA-IMS
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AGENDA

- Cluster analysis
 - Clustering techniques
 - Self-organizing maps

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Clustering

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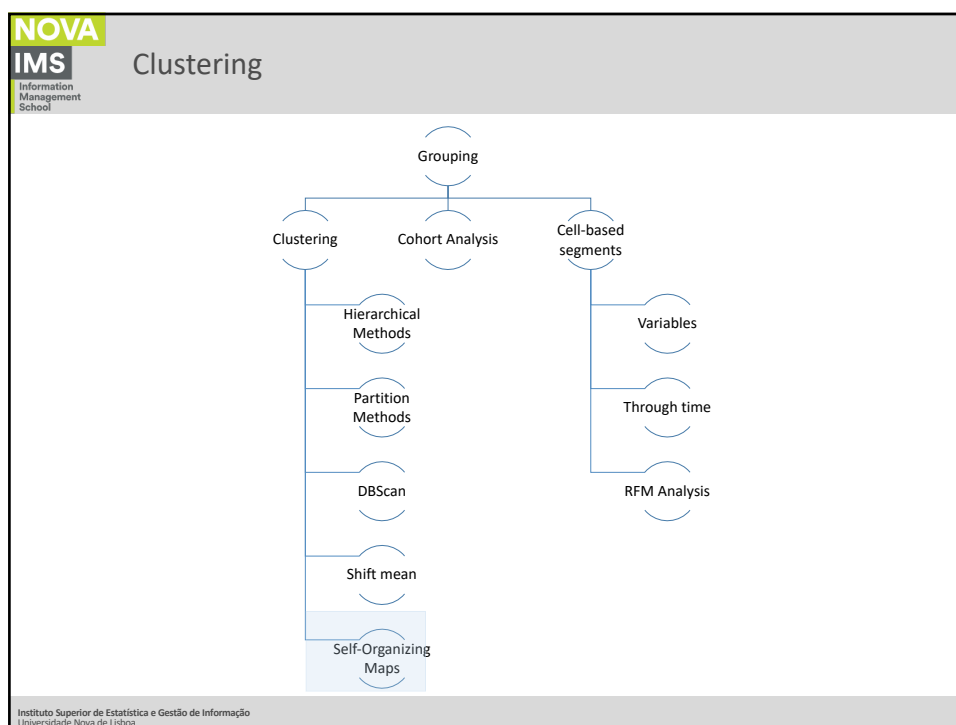








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Self-Organizing Maps

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Self-Organizing Maps

- Unsupervised neural networks;
- Closely related to clustering;
- The inputs are connected to a two-dimensional (it may have several dimensions) matrix of units (neurons);
- Each unit is connected to its neighbors.
- What is its use?
 - Multidimensional data visualization;
 - Cluster detection;
 - Market segmentation;
 - Outlier detection;
 - Solve TSP, robot control, alarm detection, etc., etc., etc.

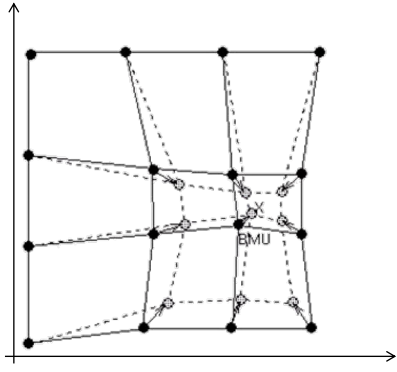
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Self-Organizing Maps

- Each neuron is a **vector in the input space**, just as the data patterns;
- During training, neurons are **pulled** to the positions of the input data, **dragging** with them their neighbors in the output space;
- The map can be seen as a **rubber sheet**, stretched and twisted, so that it passes in (or at least near) the data patterns.



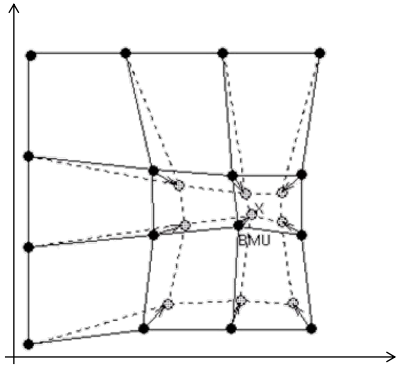
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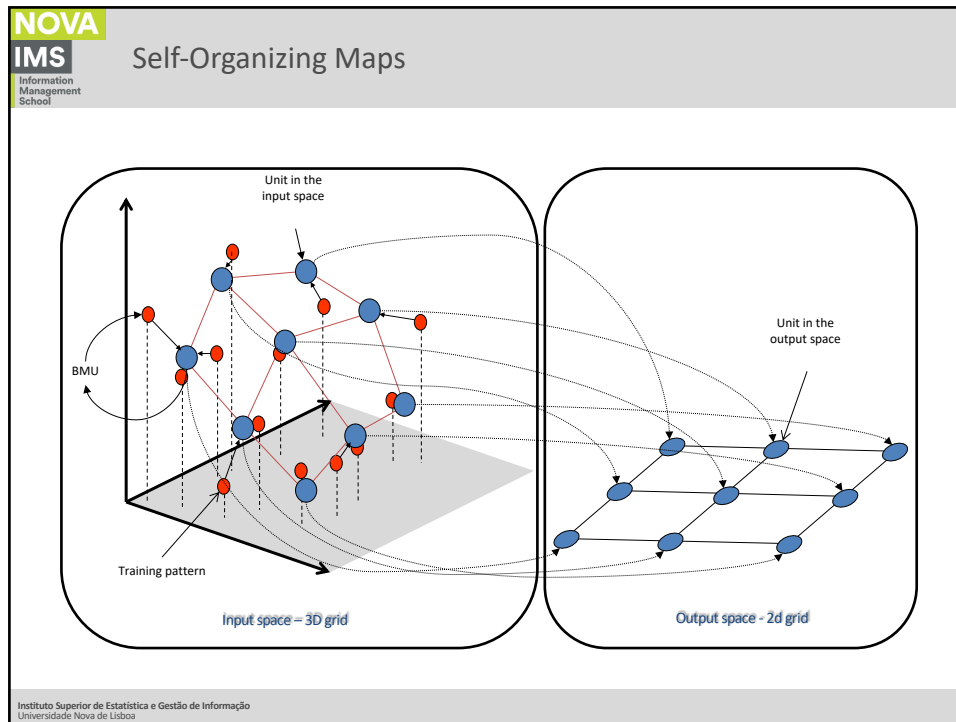
Self-Organizing Maps

- Input patterns are compared with all neurons and the **closest is considered to be the winning neuron**.
- We consider that the input pattern is **mapped to the winning neuron**.
- The **winner is updated** (so that it resembles even more the data pattern that it represents), and its neighbors are also updated a little.
- There is always a slight difference between the data and the neurons that represent them. That difference is the **quantization error**.



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Self-Organizing Maps

- **Comparisons with biology**
 - Biological systems have to use some kind of self-organization and adaptation.
 - There is evidence of:
 - Layered structure in the brain;
 - Those layers seem to spatially organize the information;
 - Similar "Concepts" are mapped to adjacent areas;
 - Experimental work with animals points to an organization similar to SOM in the visual cortex.

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The SOM Algorithm

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Self-Organizing Maps

- **SOM Algorithm:**
 - Step 0:** Randomly initialize the weights w_{ij}
 - Set the neighborhood topological parameters
 - Set the learning rate
 - Step 1:** While stop condition false, do steps 2-7
 - Step 2:** For each input vector x , do steps 3-5
 - Step 3:** For each j , execute:

$$D(j) = \sum (w_{ij} - x_i)^2$$
 - Step 4:** Find the unit that minimizes $D(j)$
 - Step 5:** For every j unit within the predefined and for all the i :

$$w_{ij}(\text{new}) = w_{ij}(\text{old}) + \alpha [x_i - w_{ij}(\text{old})]$$
 - Step 6:** Update the learning rate
 - Step 7:** Update (reduce) the radius of the topological neighborhood

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Self-Organizing Maps

- **SOM Algorithm:**
 - What happens in the Input space?
 - If we only have two variables, optimization will look like this

SOM 1-dimensional

SOM 2-dimensional

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Self-Organizing Maps

- **SOM Algorithm:**
 - What happens in the output space

Color Demo

- Suppose we want to group cells according to their rgb code (red, green and blue)
- Each individual (color) represents a particular combination of rgb intensities
- In this demo, we can see how the output space is transformed as individuals are presented to the network

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Self-Organizing Maps

- **SOM Algorithm:**
 - Key outputs of the SOM:
 - U-Matrices;
 - Component planes;
 - Hit Plots.

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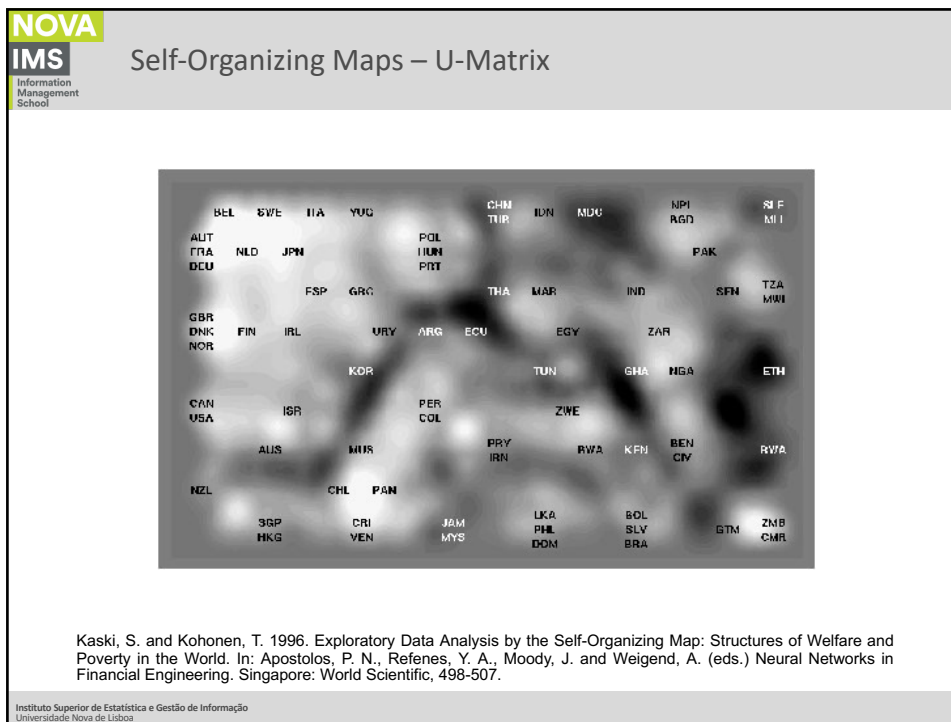
Self-Organizing Maps – U-Matrix

Input space – 3D grid

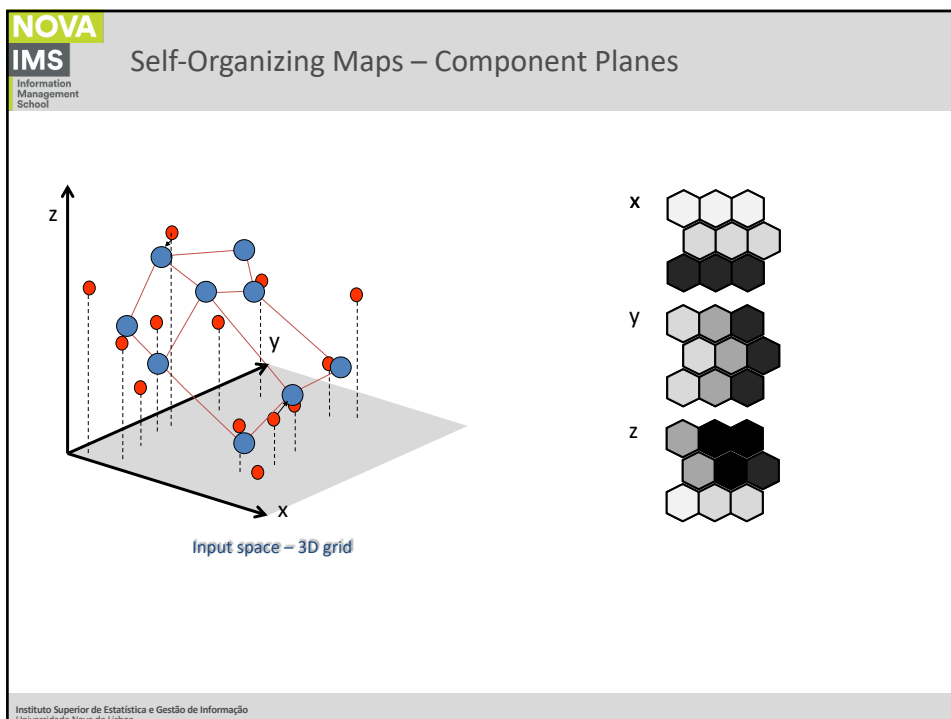
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7	•	•	31	•	5	17	•	•	•	12	•	89
98	3	46	65	•	•	1	40	9	•	28	•	79
6	•	11	74	27	72	•	•	•	•	99	•	94
58	2	10	•	26	60	•	29	•	•	102	97	87
45	41	57	30	•	64	25	•	•	•	90	•	76
53	•	49	69	•	•	•	16	•	•	85	•	76
65	73	•	•	88	•	43	•	84	•	86	•	104
•	•	•	•	•	51	•	•	•	•	•	•	•
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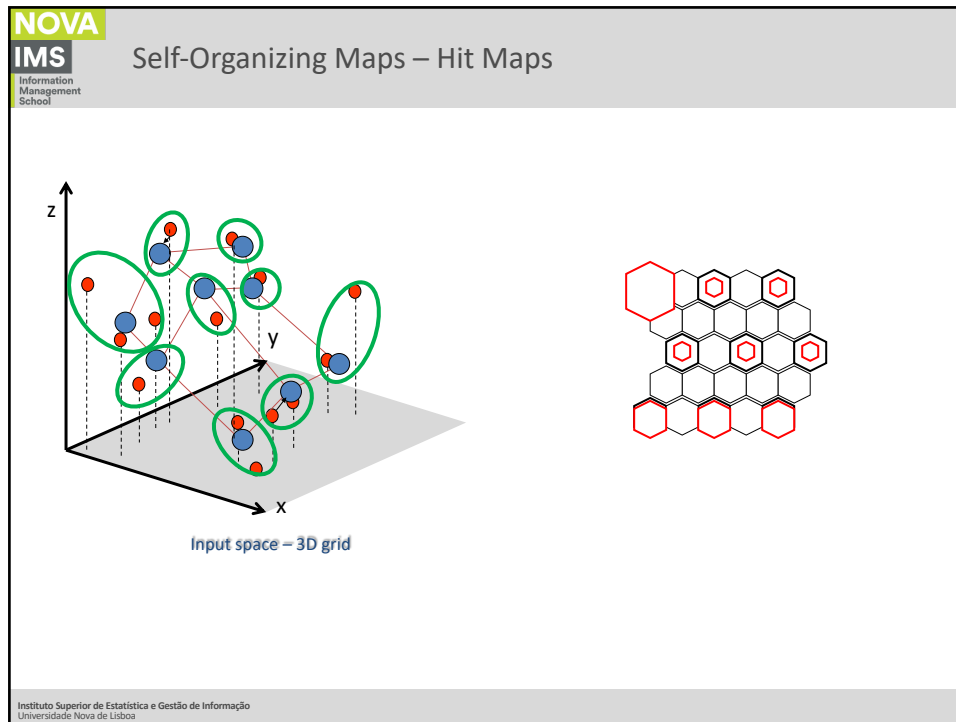
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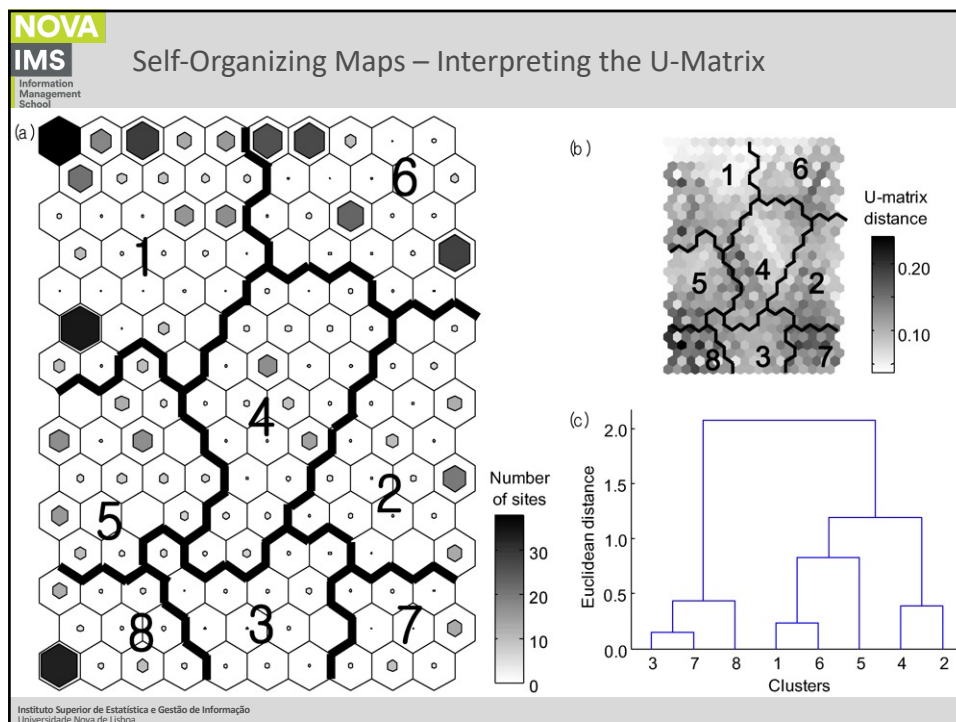
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- **SOM Algorithm:**
 - Vectors to classify:**
 $(1, 1, 0, 0); (0, 0, 0, 1); (1, 0, 0, 0); (0, 0, 1, 1)$
 - Maximum number of clusters to form:**
 $m = 2$
 - Learning rate:**
 $\alpha(0) = .6, \quad \alpha(t+1) = .5 \alpha(t)$

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- **SOM Algorithm:**

Step 0: Initial matrix of weights

(0.2, 0.6, 0.5, 0.9);

(0.8, 0.4, 0.7, 0.3);

Initial radius: $R = 0$

Initial learning rate:

$\alpha(0) = .6$

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Self-Organizing Maps

- **SOM Algorithm:**

Step 1: Initialize training

Step 2: first vector (1, 1, 0, 0);

Step 3:

$D(1) = (.2-1)^2 + (.6-1)^2 + (.5-0)^2 + (.9-0)^2$
 $= 1.86;$

$D(2) = (.8-1)^2 + (.4-1)^2 + (.7-0)^2 + (.3-0)^2$
 $= 0.98$

Vectors of weights:

(0.2, 0.6, 0.5, 0.9)

(0.8, 0.4, 0.7, 0.3)

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- **SOM Algorithm:**

Step 4: The input vector is closest to unit 2, therefore

$$j = 2 = (0.8, 0.4, 0.7, 0.3)$$

Step 5: The weights of the winning unit are adjusted

$$w_{i2}(\text{new}) = w_{i2}(\text{old}) + .6 [x_i - w_{i2}(\text{old})]$$

$$= .4 w_{i2}(\text{old}) + .6 x_i$$

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Self-Organizing Maps

- **SOM Algorithm:**

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Self-Organizing Maps

- SOM Algorithm:**

Thus, the weight matrix is adjusted:

.2	.92
.6	.76
.5	.28
.9	.12

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Self-Organizing Maps

- SOM Algorithm:**

Thus, the weight matrix is adjusted:

.08	.92
.24	.76
.20	.28
.96	.12

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